

Highlights

Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

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- SC-FMA converts knowledge-intensive traces into auditable knowledge-priority records.
- The method combines annotation fidelity, structural dependency, redundancy, and bottleneck roles.
- PRM800K evidence supports fixed-budget audit prioritization under annotation signals.
- Knowledge-graph representation checks show structural-priority sensitivity.
- Audit-target retrieval improves Recall@25% from 0.235 to 0.699 over a scalar-only view.